

Report for AI beam optimization tests

Model / Run	Step / Proposal	Width (mm)	Height (mm)	Mass (kg)	Deflection (mm)	Stress (MPa)	Feasible	valid?
llama3.2:1b (OLLAMA) - Run #1	Prop 11	90	100	N/A	N/A	N/A	N/A	INVALID REPLY
llama3.2:1b (OLLAMA) - Run #2	Iter 0 (Start)	50	80	31.4	0.488	46.9	TRUE	
llama3.2:1b (OLLAMA) - Run #2	Prop 01	80	100	N/A	N/A	N/A	N/A	INVALID REPLY
llama3.2:1b (OLLAMA) - Run #2	Prop 02	70	80	43.96	0.349	33.5	TRUE	
llama3.2:1b (OLLAMA) - Run #2	Prop 03	40	80	25.12	0.61	58.6	TRUE	
llama3.2:1b (OLLAMA) - Run #2	Prop 04	70	80	N/A	N/A	N/A	N/A	INVALID REPLY
llama3.2:1b (OLLAMA) - Run #2	Prop 05	80	100	N/A	N/A	N/A	N/A	INVALID REPLY
llama3.2:1b (OLLAMA) - Run #2	Prop 06	70	80	N/A	N/A	N/A	N/A	INVALID REPLY
llama3.2:1b (OLLAMA) - Run #2	Prop 07	70	80	N/A	N/A	N/A	N/A	INVALID REPLY
llama3.2:1b (OLLAMA) - Run #2	Prop 08	90	85	N/A	N/A	N/A	N/A	INVALID REPLY
llama3.2:1b (OLLAMA) - Run #2	Prop 09	80	100	N/A	N/A	N/A	N/A	INVALID REPLY
llama3.2:1b (OLLAMA) - Run #2	Prop 10	90	90	N/A	N/A	N/A	N/A	INVALID REPLY
llama3.2:1b (OLLAMA) - Run #2	Prop 11	80	100	N/A	N/A	N/A	N/A	INVALID REPLY
gpt-4o-mini (OPENAI) - Run #1	Iter 0 (Start)	50	80	31.4	0.488	46.9	TRUE	
gpt-4o-mini (OPENAI) - Run #1	Prop 01	50	70	27.47	0.729	61.2	TRUE	
gpt-4o-mini (OPENAI) - Run #1	Prop 02	50	60	23.55	1.157	83.3	FALSE	
gpt-4o-mini (OPENAI) - Run #1	Prop 03	50	65	25.51	0.91	71	TRUE	
gpt-4o-mini (OPENAI) - Run #1	Prop 04	50	64	25.12	0.954	73.2	TRUE	
gpt-4o-mini (OPENAI) - Run #1	Prop 05	50	63	24.73	1	75.6	TRUE	
gpt-4o-mini (OPENAI) - Run #1	Prop 06	50	62	24.33	1.049	78	FALSE	
gpt-4o-mini (OPENAI) - Run #1	Prop 07	50	61	23.94	1.101	80.6	FALSE	
gpt-4o-mini (OPENAI) - Run #1	Prop 08	50	60	23.55	1.157	83.3	FALSE	
gpt-4o-mini (OPENAI) - Run #1	Prop 09	50	63.5	24.92	0.976	74.4	TRUE	
gpt-4o-mini (OPENAI) - Run #1	Prop 10	50	63	24.73	1	75.6	TRUE	
gpt-4o-mini (OPENAI) - Run #1	Prop 11	50	62.5	24.53	1.024	76.8	FALSE	
gpt-4o-mini (OPENAI) - Run #2	Iter 0 (Start)	50	80	31.4	0.488	46.9	TRUE	
gpt-4o-mini (OPENAI) - Run #2	Prop 01	50	70	27.47	0.729	61.2	TRUE	
gpt-4o-mini (OPENAI) - Run #2	Prop 02	50	60	23.55	1.157	83.3	FALSE	
gpt-4o-mini (OPENAI) - Run #2	Prop 03	50	65	25.51	0.91	71	TRUE	
gpt-4o-mini (OPENAI) - Run #2	Prop 04	50	64	25.12	0.954	73.2	TRUE	
gpt-4o-mini (OPENAI) - Run #2	Prop 05	50	63	24.73	1	75.6	TRUE	
gpt-4o-mini (OPENAI) - Run #2	Prop 06	50	62	24.33	1.049	78	FALSE	
gpt-4o-mini (OPENAI) - Run #2	Prop 07	50	61	23.94	1.101	80.6	FALSE	
gpt-4o-mini (OPENAI) - Run #2	Prop 08	50	60	23.55	1.157	83.3	FALSE	
gpt-4o-mini (OPENAI) - Run #2	Prop 09	50	63.5	24.92	0.976	74.4	TRUE	
gpt-4o-mini (OPENAI) - Run #2	Prop 10	50	63	24.73	1	75.6	TRUE	
gpt-4o-mini (OPENAI) - Run #2	Prop 11	50	62.5	24.53	1.024	76.8	FALSE	

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=====
Running llama3.2:1b (OLLAMA) - Run #1
=====
Iter 0 (Start): width: 50.0 mm, height: 80.0 mm -> Mass: 31.40 kg, Deflection: 0.488 mm, Stress: 46.9 MPa, Feasible: True
Prop 01: INVALID REPLY (Format Error) -> '70.0 mm, 100.0 mm'
Prop 02: width: 80.0 mm, height: 100.0 mm -> Mass: 62.80 kg, Deflection: 0.156 mm, Stress: 18.8 MPa, Feasible: True
Prop 03: INVALID REPLY (Format Error) -> '70.0 mm, 100.0 mm'
Prop 04: width: 85.0 mm, height: 100.0 mm -> Mass: 66.72 kg, Deflection: 0.147 mm, Stress: 17.6 MPa, Feasible: True
Prop 05: INVALID REPLY (Format Error) -> '80.0 mm, 100.0 mm'
Prop 06: INVALID REPLY (Format Error) -> '70.0 mm, 90.0 mm'
Prop 07: INVALID REPLY (Format Error) -> '80.0 mm, 100.0 mm'
Prop 08: INVALID REPLY (Format Error) -> '80.0 mm, 100.0 mm'
Prop 09: INVALID REPLY (Format Error) -> '90.0 mm, 100.0 mm'
Prop 10: width: 90.0 mm, height: 100.0 mm -> Mass: 70.65 kg, Deflection: 0.139 mm, Stress: 16.7 MPa, Feasible: True
Prop 11: INVALID REPLY (Format Error) -> '90.0 mm, 100.0 mm'

=====
Running llama3.2:1b (OLLAMA) - Run #2
=====
Iter 0 (Start): width: 50.0 mm, height: 80.0 mm -> Mass: 31.40 kg, Deflection: 0.488 mm, Stress: 46.9 MPa, Feasible: True
Prop 01: INVALID REPLY (Format Error) -> '80.0 mm, 100.0 mm'
Prop 02: width: 70.0 mm, height: 80.0 mm -> Mass: 43.96 kg, Deflection: 0.349 mm, Stress: 33.5 MPa, Feasible: True
Prop 03: width: 40.0 mm, height: 80.0 mm -> Mass: 25.12 kg, Deflection: 0.610 mm, Stress: 58.6 MPa, Feasible: True
Prop 04: INVALID REPLY (Format Error) -> '70.0 mm, 80.0 mm'
Prop 05: INVALID REPLY (Format Error) -> '80.0 mm, 100.0 mm'
Prop 06: INVALID REPLY (Format Error) -> '70.0 mm, 80.0 mm'
Prop 07: INVALID REPLY (Format Error) -> '70.0 mm, 80.0 mm'
Prop 08: INVALID REPLY (Format Error) -> '90.0 mm, 85.0 mm'
Prop 09: INVALID REPLY (Format Error) -> '80.0 mm, 100.0 mm'
Prop 10: INVALID REPLY (Format Error) -> '90.0 mm, 90.0 mm'
Prop 11: INVALID REPLY (Format Error) -> '80.0 mm, 100.0 mm'

=====
Running gpt-4o-mini (OPENAI) - Run #1
=====
Iter 0 (Start): width: 50.0 mm, height: 80.0 mm -> Mass: 31.40 kg, Deflection: 0.488 mm, Stress: 46.9 MPa, Feasible: True
Prop 01: width: 50.0 mm, height: 70.0 mm -> Mass: 27.47 kg, Deflection: 0.729 mm, Stress: 61.2 MPa, Feasible: True
Prop 02: width: 50.0 mm, height: 60.0 mm -> Mass: 23.55 kg, Deflection: 1.157 mm, Stress: 83.3 MPa, Feasible: False
Prop 03: width: 50.0 mm, height: 65.0 mm -> Mass: 25.51 kg, Deflection: 0.910 mm, Stress: 71.0 MPa, Feasible: True
Prop 04: width: 50.0 mm, height: 64.0 mm -> Mass: 25.12 kg, Deflection: 0.954 mm, Stress: 73.2 MPa, Feasible: True
Prop 05: width: 50.0 mm, height: 63.0 mm -> Mass: 24.73 kg, Deflection: 1.000 mm, Stress: 75.6 MPa, Feasible: True
Prop 06: width: 50.0 mm, height: 62.0 mm -> Mass: 24.33 kg, Deflection: 1.049 mm, Stress: 78.0 MPa, Feasible: False
Prop 07: width: 50.0 mm, height: 61.0 mm -> Mass: 23.94 kg, Deflection: 1.101 mm, Stress: 80.6 MPa, Feasible: False
Prop 08: width: 50.0 mm, height: 60.0 mm -> Mass: 23.55 kg, Deflection: 1.157 mm, Stress: 83.3 MPa, Feasible: False
Prop 09: width: 50.0 mm, height: 63.5 mm -> Mass: 24.92 kg, Deflection: 0.976 mm, Stress: 74.4 MPa, Feasible: True
Prop 10: width: 50.0 mm, height: 63.0 mm -> Mass: 24.73 kg, Deflection: 1.000 mm, Stress: 75.6 MPa, Feasible: True
Prop 11: width: 50.0 mm, height: 62.5 mm -> Mass: 24.53 kg, Deflection: 1.024 mm, Stress: 76.8 MPa, Feasible: False

=====
Running gpt-4o-mini (OPENAI) - Run #2
=====
Iter 0 (Start): width: 50.0 mm, height: 80.0 mm -> Mass: 31.40 kg, Deflection: 0.488 mm, Stress: 46.9 MPa, Feasible: True
Prop 01: width: 50.0 mm, height: 70.0 mm -> Mass: 27.47 kg, Deflection: 0.729 mm, Stress: 61.2 MPa, Feasible: True
Prop 02: width: 50.0 mm, height: 60.0 mm -> Mass: 23.55 kg, Deflection: 1.157 mm, Stress: 83.3 MPa, Feasible: False
Prop 03: width: 50.0 mm, height: 65.0 mm -> Mass: 25.51 kg, Deflection: 0.910 mm, Stress: 71.0 MPa, Feasible: True
Prop 04: width: 50.0 mm, height: 64.0 mm -> Mass: 25.12 kg, Deflection: 0.954 mm, Stress: 73.2 MPa, Feasible: True
Prop 05: width: 50.0 mm, height: 63.0 mm -> Mass: 24.73 kg, Deflection: 1.000 mm, Stress: 75.6 MPa, Feasible: True
Prop 06: width: 50.0 mm, height: 62.0 mm -> Mass: 24.33 kg, Deflection: 1.049 mm, Stress: 78.0 MPa, Feasible: False
Prop 07: width: 50.0 mm, height: 61.0 mm -> Mass: 23.94 kg, Deflection: 1.101 mm, Stress: 80.6 MPa, Feasible: False
Prop 08: width: 50.0 mm, height: 60.0 mm -> Mass: 23.55 kg, Deflection: 1.157 mm, Stress: 83.3 MPa, Feasible: False
Prop 09: width: 50.0 mm, height: 63.5 mm -> Mass: 24.92 kg, Deflection: 0.976 mm, Stress: 74.4 MPa, Feasible: True
Prop 10: width: 50.0 mm, height: 63.0 mm -> Mass: 24.73 kg, Deflection: 1.000 mm, Stress: 75.6 MPa, Feasible: True
Prop 11: width: 50.0 mm, height: 62.5 mm -> Mass: 24.53 kg, Deflection: 1.024 mm, Stress: 76.8 MPa, Feasible: False

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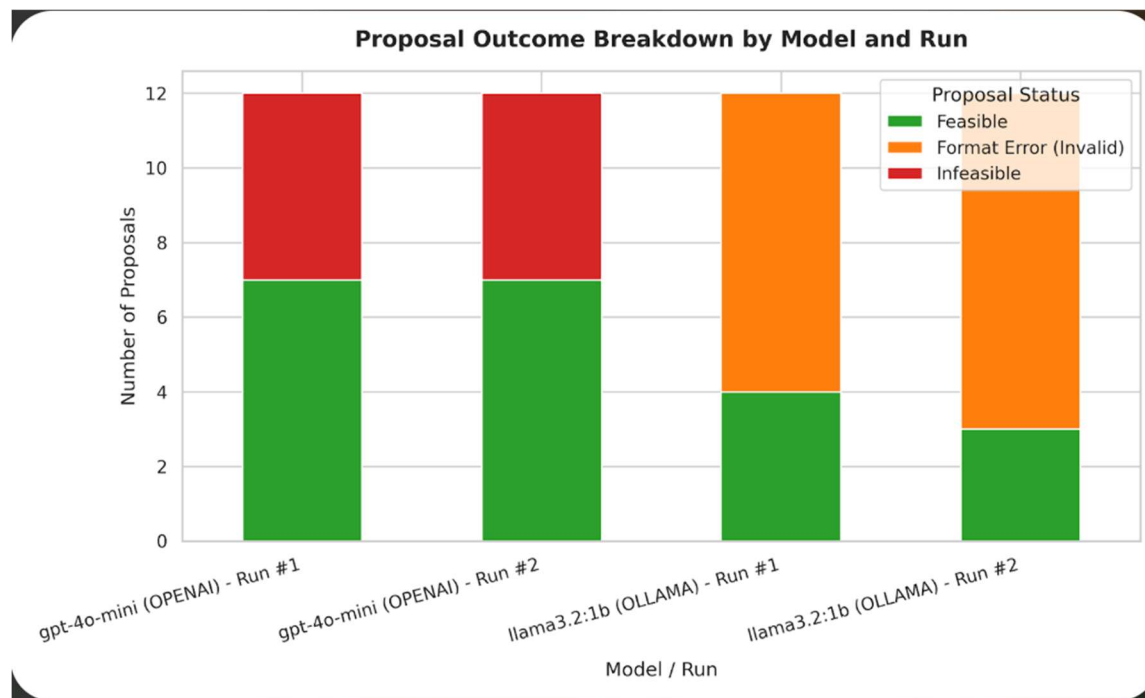
Results spreadsheet and actual output

Comparisons and limitations of using AI for the optimizations:

The difference between ollama and openai is astounding as in how inconsistent ollama is. Despite it giving the same hypothetical width and height, it thinks that the dimensions are out of range despite them not being out of range and ollama also JUST calculated it last iteration. Openai is significantly more consistent and correct majority of the time, however it didn't provide very varied results and just oscillated between 60-65 mm for height while keeping width the same.

While openai did do quite a decent job making calculations and doing what the evaluate beam function did, when trying to propose a new width and height, it wasn't very different and often gave very similar dimensions.

Graph of amount of “feasible” proposals given by the AIs:



Now ollama is a bit funky because half of its feasible results are just wrong